

Software Requirements Specification

AI-Powered Interview Preparation Platform

Version 1.0 — Consolidated Final Scope

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1. Introduction

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for the AI-Powered Interview Preparation Platform, a web-based system that allows job seekers to practice realistic, role-specific mock interviews with an AI interviewer and receive structured, post-session feedback. This document is the single, consolidated specification for the system. It is intended to guide the design, development, and testing of the system, and to serve as an agreement between the project stakeholders on what the system will and will not do.

Job seekers frequently struggle to prepare effectively for interviews because traditional methods — reading generic guides, watching videos, or arranging mock interviews with friends — are inconsistent, do not scale, and rarely simulate real interview pressure or provide objective, data-driven feedback on communication, technical accuracy, and confidence. This platform addresses that gap by combining conversational AI, speech processing, and structured evaluation frameworks such as STAR, and — building on the baseline capability set — closes several gaps identified against competitor products (Huru.ai, Final Round AI, Yoodli, Interviews.Chat, and SpeakPrep AI), notably content/relevance scoring, adaptive follow-up questioning, job-description-driven personalization, model-answer generation, and optional video-based body-language and presence feedback, while preserving the platform's unique, unmatched capability of longitudinal real-world outcome tracking.

2. Purpose

The purpose of this document is to define, in unambiguous terms, the final scope, features, and constraints of the AI-Powered Interview Preparation Platform. It is intended for use by the developer(s) implementing the system, the academic evaluators/instructors assessing the project, and any future contributor who needs a single, precise reference for what the system is expected to deliver — and, equally importantly, what it will deliberately continue to exclude. This document consolidates and replaces the two prior draft SRS versions; where those drafts differed, this document reflects the resolved, final decision.

3. Scope

The system is a single-user, web-based interview preparation tool covering Technical, Behavioral, and HR interview rounds. It is explicitly a practice-only tool and is scoped as follows:

- Voice-based mock interview practice using speech-to-text (English only).
- Role-based, resume-based, and optional job-description-based personalization of interview questions, so generated questions can be tailored to a specific company/posting rather than a generic role.
- Adaptive, conversational follow-up questions generated in response to the candidate's actual answers, layered on top of a pre-generated core question sequence, with no feedback or scoring shown mid-session, to preserve realistic interview pressure.

- A comprehensive post-interview feedback report covering delivery (pace, filler words, tone), STAR-method structure for behavioral answers, content/relevance scoring of answer substance, and a rewritten model/sample answer for each question.
- Optional video capture with body-language and presence feedback (eye contact, posture, nervous habits), subject to user consent and a privacy review.
- Logging of real-world interview outcomes (offer/rejection, round reached) to enrich the user's profile for future personalization — a capability with no direct equivalent among reviewed competitor products, and the platform's core differentiator.
- A persistent, cross-session progress dashboard and skill portfolio, with explicit per-metric trend charts (pace, filler words, STAR score, content score) tracked over time.
- User authentication to associate resumes, job descriptions, sessions, and progress data with an individual account.
- Containerized deployment using Docker.

Explicit Scope Boundary — Live Interview Assistance: this platform is, and will remain, a prep-only tool. It does not, and will not, provide real-time answer suggestions, coaching, or any other form of assistance during an actual live interview call. Several competitors (e.g., Final Round AI, Interviews.Chat) offer a "copilot"-style real-time assistance mode; this platform deliberately does not, since such tools raise integrity concerns by enabling deception in real interviews. This exclusion is a permanent ethical design decision and a stated trust differentiator, not a temporary limitation of the current phase.

Out of scope for this version: live coding round simulation, multi-language support, panel/group interview simulation, and organization/B2B accounts.

4. Definitions, Acronyms and Abbreviations

Term	Definition
STT	Speech-to-Text — the process of converting spoken audio into a text transcript.
TTS	Text-to-Speech — the process of converting text into synthesized spoken audio, used for the AI interviewer's voice.
STAR	Situation, Task, Action, Result — a structured framework for answering behavioral interview questions.
NLP	Natural Language Processing — techniques used to analyze and evaluate the text transcript of a user's answer.
LLM	Large Language Model — the AI model used to generate interview questions, contextual follow-ups, content scoring, and model answers.
JD	Job Description — the text of a specific job posting, used to further tailor generated interview questions beyond role and resume alone.
Content/Relevance Score	A rating of whether the substance of a candidate's answer is accurate and pertinent to the question asked, as distinct from delivery-style metrics such as pace or filler words.
SRS	Software Requirements Specification — this document.

UI	User Interface.
API	Application Programming Interface.

5. Overall Description

5.1 Product Perspective

The platform is a new, standalone, self-contained web product. It is not a component of an existing system, though it integrates with third-party services for speech processing, AI question generation/evaluation, authentication, and hosting. The system follows a client-server architecture with a React.js single-page front end, a Node.js back end, and a hybrid relational/NoSQL data layer, including a job-description ingestion path and an AI-evaluation pipeline covering content scoring, model-answer generation, and adaptive follow-ups.

5.2 Product Functions (Summary)

- Role, interview-type, resume, and job-description-based question personalization.
- AI-generated interview questions with adaptive, context-aware follow-ups based on the candidate's live answers.
- Voice capture and speech-to-text conversion of the candidate's spoken answers.
- Optional video capture and body-language/presence analysis (eye contact, posture, nervous habits) during practice sessions, subject to user consent.
- Post-interview delivery analysis (pace, filler words, tone) and STAR-method compliance checking.
- Post-interview content/relevance scoring of answer substance.
- Model/sample answer generation for each question as a learning reference.
- Consolidated, structured feedback report generation, including trend-aware per-question scorecards.
- Post-interview real-world outcome logging that feeds back into future personalization.
- Cross-session, cross-role progress dashboard with explicit per-metric trend charts over time.
- Secure user registration, login, and session management.

5.3 Operating Environment

The system operates as a browser-based web application accessible on modern desktop and laptop browsers with microphone access (Chrome, Edge, Firefox — latest two major versions). The back end and databases are hosted on a local host for the development phase. In future deployment phases they will be on a cloud platform (AWS or GCP) with services containerized via Docker for consistent deployment across environments.

5.4 Design and Implementation Constraints

The system depends on third-party LLM, speech-to-text, and text-to-speech APIs for its core AI functionality, including content-scoring and model-answer-generation; behavior, quality, and cost are

bound by those providers' terms, latency, and rate limits. Adaptive follow-up generation introduces additional LLM calls per answer and must be designed to stay within acceptable session latency. Voice interaction is supported in English only in this version. The application is single-tenant/single-user in this phase, with no organizational or multi-tenant data model.

5.5 User Documentation

The delivered system will include in-app onboarding guidance (tooltips/first-run walkthrough) covering resume and job-description upload, starting a mock session, and reading the feedback report (including model answers and trend charts), in place of a separate user manual.

6. User Characteristics

The system is designed for three primary user groups:

- Final-year students / freshers preparing for campus placements — assumed to have basic computer literacy and limited real interview experience; likely to rely heavily on model answers and STAR guidance.
- Job seekers switching careers or industries — assumed to have some prior interview experience but require reframing of past experience for a new domain and role/JD-specific question relevance.
- Working professionals preparing for promotions or new roles — assumed to have strong domain expertise but limited recent interview practice; likely to value content/relevance scoring and trend tracking most.

All user groups are assumed to have conversational proficiency in spoken and written English, access to a device with a working microphone, and a stable internet connection, and are expected to have access to a target job description when they wish to use JD-based personalization (optional where unavailable). No specialized technical training is required to use the platform; the interface is designed for general computer-literate users, not exclusively for software engineers.

7. Functional Requirements

Each functional requirement is identified by an ID of the form R.x, decomposed into sub-requirements R.x.y, and specified in terms of Input, Output, and Processing.

R.1 Interview Setup (Role, Interview Type, Resume Upload, Job Description)

Allows the user to select a job role and interview type, upload a resume, and optionally provide a job description; the system parses these inputs and configures the interview session accordingly.

R.1.1 Select Role and Interview Type

Input	Users select a role (e.g., Software Engineer, Data Analyst) and an interview type (Technical, HR, Behavioral) from a dropdown/list.
Output	Selected role and interview type are set for the session.

Processing	System stores the selection and configures the relevant question bank.
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R.1.2 Upload Resume

Input	User uploads a resume file (PDF/DOCX).
Output	Confirmation that the resume has been uploaded and parsed.
Processing	System extracts text and key details (skills, experience, projects) using a parsing engine.

R.1.3 Enter Job Description

Input	User optionally pastes or uploads the text of a target job description.
Output	Confirmation that the job description has been captured and linked to the session.
Processing	System extracts key requirements, keywords, and responsibilities from the job description text for use in question generation.

R.1.4 Generate Interview Interface

Input	Selected role, interview type, parsed resume data, and (if provided) parsed job-description data.
Output	Interview interface is launched with the first question displayed.
Processing	System generates contextual questions based on role, interview type, resume content, and job-description content when available.

R.2 User Answer Management

Captures the user's spoken answer to each question, converts it to text, generates adaptive follow-up questions where appropriate, and displays the transcript back to the user.

R.2.1 Capture Answer

Input	User's spoken answer (audio).
Output	Recorded audio answer.
Processing	System records audio input via the microphone.

R.2.2 Convert Speech to Text

Input	Recorded audio answer.
Output	Text transcript of the answer displayed to the user.
Processing	Speech-to-text engine converts audio into text.

R.2.3 Generate Adaptive Follow-up Question

Input	Text transcript of the candidate's most recent answer and current question context.
Output	A contextual follow-up question, when the candidate's answer warrants one, inserted into the session flow.
Processing	System evaluates the answer transcript against the current line of questioning and uses the LLM to generate a targeted follow-up question, or proceeds to the next planned question if no follow-up is warranted.

R.2.4 Capture Video (Optional)

Input	User enables the camera toggle and provides consent for video capture.
Output	Video recorded alongside audio for the current answer.
Processing	System captures a video stream via the browser's camera API, subject to user consent.

R.2.5 Proceed to Next Question

Input	Click on "Next" button.
Output	Next question (planned or adaptively generated) is displayed.
Processing	System fetches the next question from the configured question set or the adaptively generated follow-up.

R.3 Feedback Report

Analyzes the user's answers at the end of the interview and generates an evaluation report covering delivery, STAR-method compliance, content/relevance, model answers, body-language and presence (when available), compiled into a single per-question scorecard.

R.3.1 Analyze Answer Delivery

Input	User's recorded answers (audio and text transcript).
Output	Calculated metrics for pace, filler word count, and tone.
Processing	Audio/NLP analysis engine evaluates speech pace, detects filler words, and assesses tone.

R.3.2 Verify STAR Method

Input	Text transcript of the answer.
Output	STAR (Situation, Task, Action, Result) compliance rating and suggestions.
Processing	System checks the structure of the answer against the STAR framework.

R.3.3 Evaluate Content Relevance and Correctness

Input	Text transcript of the answer, the original question, and (if available) resume/job-description context.
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Output	A content/relevance score and qualitative notes on factual accuracy and relevance.
Processing	LLM-based evaluation compares the answer's substance against the question intent and available context to assess relevance and correctness.

R.3.4 Generate Model/Sample Answer

Input	The original question, the candidate's transcript, and relevant resume/job-description context.
Output	A rewritten, improved example answer for the same question.
Processing	System prompts the LLM to produce a concise, well-structured model answer that the candidate can use as a learning reference.

R.3.5 Analyze Body Language and Presence

Input	Recorded video clip (if video capture was enabled under R.2.4).
Output	Body-language and presence metrics (eye contact, posture, nervous habits) included in the feedback report.
Processing	System analyzes sampled frames from the recorded video to assess eye contact, posture, and nervous habits, subject to a separate privacy and consent review.

R.3.6 Generate Evaluation Report

Input	Aggregated analysis from R.3.1–R.3.5.
Output	Evaluation report displayed to the user as per-question scorecards, showing delivery scores, STAR rating, content/relevance score, model answer, body-language/presence metrics (when captured), and improvement suggestions.
Processing	System compiles all analyzed data into a structured, per-question report format.

R.4 Post-Interview Outcome

Asks the user about a recent real-world interview they attended and adds that context to the user's profile for future personalization. This capability has no direct equivalent among reviewed competitor products and remains the platform's core differentiator.

R.4.1 Fill Post-Interview Form

Input	User fills a form with interview-specific fields (company name, role, round, outcome, difficulty).
Output	Submission confirmation shown to the user.
Processing	System validates the submitted form data.

R.4.2 Generate Summary and Update Profile

Input	Submitted form data.
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Output	User profile updated with the new interview context.
Processing	System summarizes the form fields and stores them in the user's profile for use in future question personalization.

R.5 Progress Dashboard

Shows the user's progress across past interview sessions, fetched from the database, with explicit trend visualization per metric.

R.5.1 Fetch Progress Data

Input	User accesses the dashboard (dashboard load request).
Output	Retrieved historical performance data for the user, including per-metric score history.
Processing	System queries the database for the user's past interview sessions and per-metric scores (pace, filler words, STAR, content/relevance).

R.5.2 Display Progress Trends

Input	Retrieved progress data.
Output	Visual display of progress including explicit line/trend charts per metric across sessions, alongside a summary of strengths/weaknesses over time.
Processing	System renders the dashboard UI with per-metric trend charts (using Chart.js/Recharts) plus a summary snapshot (total sessions, average score, streak).

R.6 User Authentication

Authenticates users so that resumes, job descriptions, interview sessions, feedback reports, and progress data can be associated with a specific profile. This is a prerequisite for R.4 (profile updates) and R.5 (progress dashboard).

R.6.1 Register

Input	User's name, email, password (or OAuth credentials).
Output	New user account created; confirmation shown.
Processing	System validates input and creates a user record in the database.

R.6.2 Login

Input	Registered email and password (or OAuth login).
Output	User is authenticated and redirected to the home/dashboard interface.
Processing	System verifies credentials against stored records and creates a session.

R.6.3 Logout

Input	Click on "Logout" button.
Output	User session is ended; user is redirected to the login screen.
Processing	System terminates the active session.

8. Non-functional Requirements

Category	Requirement
Performance	Speech-to-text display latency shall remain under 3 seconds per answer despite the adaptive follow-up and content-scoring pipelines. Question generation shall complete within 5 seconds of session start, and adaptive follow-up generation shall resolve within 4 seconds of answer submission so the session flow feels continuous.
Usability	The setup flow (role/type selection + resume upload + optional job-description entry) shall be completable by a first-time user in under 4 steps without external help.
Reliability	Audio capture and transcripts shall not be lost or corrupted during an active session; the system shall auto-save transcripts as they are generated, and adaptive follow-up generation failures shall gracefully fall back to the next planned question rather than blocking the session.
Availability	The platform shall target 99% uptime for the web application and API layer, excluding scheduled maintenance.
Security	All user data (resumes, job descriptions, transcripts, credentials) shall be encrypted in transit (TLS) and at rest. Passwords shall be hashed; OAuth tokens shall never be stored in plain text.
Scalability	The architecture shall support horizontal scaling of the backend and AI-processing services (including content-scoring and model-answer generation load) via containerization to accommodate growth in concurrent sessions.
Maintainability	The codebase shall follow a modular service structure (setup, session, adaptive-questioning, feedback/scoring, dashboard, auth) so that individual modules can be updated independently.
Portability	The containerized backend shall be deployable on any Docker-compatible cloud environment (AWS or GCP) without code changes.
Ethical / Trust Boundary	The system shall not provide, and shall have no code path enabling, real-time answer assistance during an actual live interview call. This constraint shall be verifiable in the architecture (no live-call integration exists) and stated explicitly in user-facing materials.

9. Constraints

- The system is single-user (individual candidate) only; no multi-tenant or organization/admin features exist in this phase.

- Voice interaction is supported in English only.
- Live coding round simulation is explicitly out of scope.
- The system shall never provide real-time assistance during an actual live interview; feedback, model answers, and adaptive follow-ups are only ever generated within a practice session, never during a real interview call. This is a deliberate and permanent ethical boundary.
- The system's AI features (question generation, adaptive follow-ups, content scoring, model answers) are dependent on the availability, accuracy, cost, and rate limits of third-party LLM, STT, and TTS providers.
- A stable internet connection and functioning microphone are required on the client device; the system does not support offline use.
- Job-description input is optional; the system must still function using role and resume alone when no job description is supplied.

10. Assumptions

- Users have access to a device with a working microphone and a modern web browser.
- Users upload resumes, and optionally job descriptions, that accurately reflect their skills, experience, and the role they are targeting.
- The chosen LLM API is capable of generating role- and JD-relevant, contextually appropriate, unbiased interview questions, adaptive follow-ups, content/relevance scores, and model answers.
- Users voluntarily and honestly report real-world interview outcomes in R.4; the system does not independently verify these outcomes.
- Third-party speech-to-text accuracy is sufficient for standard, clearly spoken English at typical interview pacing.
- LLM-generated content scores and model answers, while helpful references, are treated by users as guidance rather than an authoritative or infallible assessment.

11. Software Requirements

Category	Requirement
Frontend	React.js (single-page application).
Backend	Node.js (REST/API service layer).
AI / LLM	LLM API for resume and job-description parsing, question generation, adaptive follow-up generation, content/relevance scoring, and model-answer generation.
Speech Processing	Deepgram for speech-to-text; a text-to-speech engine for the AI interviewer's voice.
Video / Body-Language Analysis	Computer-vision based analysis engine for eye contact, posture, and presence feedback from optional session video.
Database	PostgreSQL for structured data (profiles, scores, outcomes, per-metric history); MongoDB for session/transcript data.

Hosting	AWS or GCP (cloud infrastructure).
Analytics / Dashboard	Chart.js or Recharts for per-metric trend visualization.
Containerization	Docker for packaging and deployment of backend services.
Supported Browsers	Latest two major versions of Chrome, Edge, and Firefox with microphone (WebRTC) support.

12. Hardware Requirements

12.1 Client Side

- A desktop, laptop, or comparable device with a modern web browser.
- A functioning microphone for voice input and speakers/headphones for audio output.
- A functioning webcam, required only if the optional video capture and body-language analysis feature is enabled.
- A minimum internet connection of 1 Mbps upload/download for stable voice streaming.

12.2 Server Side

- Cloud-hosted compute instances (AWS or GCP) sized to support containerized backend services and the additional AI-processing workloads (adaptive follow-ups, content scoring, model-answer generation, and body-language analysis).
- Scalable storage for PostgreSQL (structured data, including expanded per-metric history) and MongoDB (session/transcript data).
- Sufficient outbound bandwidth to call third-party LLM, STT, and TTS APIs with acceptable latency, accounting for the higher call volume introduced by adaptive follow-ups and content scoring.

13. User Interface Requirements

- Setup Screen: dropdowns for role and interview type, a resume upload control, and an optional job-description text/upload field, with a single-screen onboarding form and an upload/parse confirmation state.
- Interview Session Screen: a question/avatar panel, a live-updating transcript area, a microphone/recording indicator, an elapsed-time indicator, and a "Next" control — with no feedback or scoring shown during the session, even as adaptive follow-ups are inserted.
- Optional Camera Preview: a small live camera preview with an enable/disable toggle, shown when video capture is active for body-language analysis.
- Feedback Report Screen: a tabbed or accordion, per-question scorecard layout, with one expandable card per question showing delivery metrics (pace, filler words, tone), STAR compliance, content/relevance score, body-language/presence metrics (when captured), and a model answer, so users can jump directly to their weakest answers.
- Post-Interview Outcome Form: fields for company name, role, interview round, outcome, and difficulty, with a clear submission confirmation.

- Progress Dashboard: a card-based summary (total sessions, average score, streak) at the top, with per-metric trend/line charts (pace, filler words, STAR score, content score) below, tracked across sessions, roles, and companies.
- Authentication Screens: registration, login (including OAuth option), and logout controls, consistent with standard web authentication patterns.

14. Future Enhancements

The following are candidate enhancements for future phases. None of these — and in particular, no form of real-time assistance during an actual live interview — are part of the current planned scope:

- Career coach / bootcamp (B2B) add-on with organization accounts, an admin dashboard, and cohort-level tracking.
- Multi-language support for voice-based practice beyond English.
- Coding-round simulation for technical interviews.
- Panel/group interview simulation with multiple interviewer personas, interruptions, and good-cop/bad-cop dynamics.